Problem 1: Naive Bayes parameter estimates

Define $n_y = \sum_{i=1}^{n} \mathbf{1}\{y_i = y\}$ and $S_{y,d} = \sum_{i:y_i=y} x_{i,d}$. The Gamma prior with shape 2 and rate 1 is

$$p(\lambda_{y,d}) = \lambda_{y,d} e^{-\lambda_{y,d}}, \qquad \lambda_{y,d} > 0.$$

The rate and scale conventions coincide here because the second parameter is 1. The naive assumption is conditional independence of the feature dimensions given the class.

(a) Estimate of π

The Bernoulli terms are the only terms in the objective depending on π:

$$\begin{aligned}\mathcal{J}_\pi(\pi) &= \sum_i [y_i \log \pi + (1 - y_i) \log(1 - \pi)] \\ &= n_1 \log \pi + n_0 \log(1 - \pi).\end{aligned}$$

For $n_0, n_1 > 0$, differentiation gives

$$\begin{aligned}\frac{d\mathcal{J}_\pi}{d\pi} &= \frac{n_1}{\pi} - \frac{n_0}{1 - \pi} = 0, \\ n_1(1 - \pi) &= n_0 \pi, \\ \widehat{\pi} &= \frac{n_1}{n_0 + n_1} = \frac{1}{n} \sum_i y_i.\end{aligned}$$

This is a maximum since

$$\frac{d^2 \mathcal{J}_\pi}{d\pi^2} = -\frac{n_1}{\pi^2} - \frac{n_0}{(1 - \pi)^2} < 0.$$

If all observations have the same label, the maximizing boundary is $\widehat{\pi} = 0$ or 1.

(b) Estimate of an arbitrary $\lambda_{y,d}$

Fix $y \in \{0, 1\}$ and dimension d. The terms depending on $\lambda_{y,d}$ are

$$\begin{aligned}\mathcal{J}_{y,d}(\lambda_{y,d}) &= \log p(\lambda_{y,d}) + \sum_{i:y_i=y} \log p(x_{i,d} \mid \lambda_{y,d}) \\ &= \log \lambda_{y,d} - \lambda_{y,d} + \sum_{i:y_i=y} [x_{i,d} \log \lambda_{y,d} - \lambda_{y,d} - \log(x_{i,d}!)] \\ &= (S_{y,d} + 1) \log \lambda_{y,d} - (n_y + 1)\lambda_{y,d} + C.\end{aligned}$$

Consequently,

$$\frac{\partial \mathcal{J}_{y,d}}{\partial \lambda_{y,d}} = \frac{S_{y,d} + 1}{\lambda_{y,d}} - (n_y + 1) = 0 \quad \Longrightarrow \quad \widehat{\lambda}_{y,d} = \frac{S_{y,d} + 1}{n_y + 1}.$$

The second derivative, $-(S_{y,d} + 1)/\lambda_{y,d}^2$, is strictly negative, so this is the unique maximum. It is the mode of the posterior Gamma$(S_{y,d} + 2, n_y + 1)$ rather than its mean. The positive prior contribution prevents zero estimated rates for unseen features.

Problem 2: Email classification

The supplied data contain 4,600 emails with 54 features. Values are used as provided, without rescaling. A random permutation with seed 4720 partitions the rows into ten equal groups of 460. Each group is held out once; each training set contains 4,140 rows. Both methods use the same randomly generated folds.

(a) Naive Bayes test results

Using the training estimates from Problem 1, score each test email by

$$s_y(x) = y \log \widehat{\pi} + (1 - y) \log(1 - \widehat{\pi}) + \sum_{d=1}^{54} [x_d \log \widehat{\lambda}_{y,d} - \widehat{\lambda}_{y,d}].$$

Predict the class with larger score. The terms $-\log(x_d!)$ cancel between classes.

Ground truth	Predicted 0	Predicted 1
0 (non-spam)	2296	491
1 (spam)	99	1714

$$\text{Accuracy} = \frac{2296 + 1714}{4600} = 87.1739\%.$$

Rows are ground truth and columns are predictions. The four entries sum to 4,600.

(b) Mean Poisson parameters

For each dimension and class, take the arithmetic average of the ten training-run rate estimates, for plotting only.

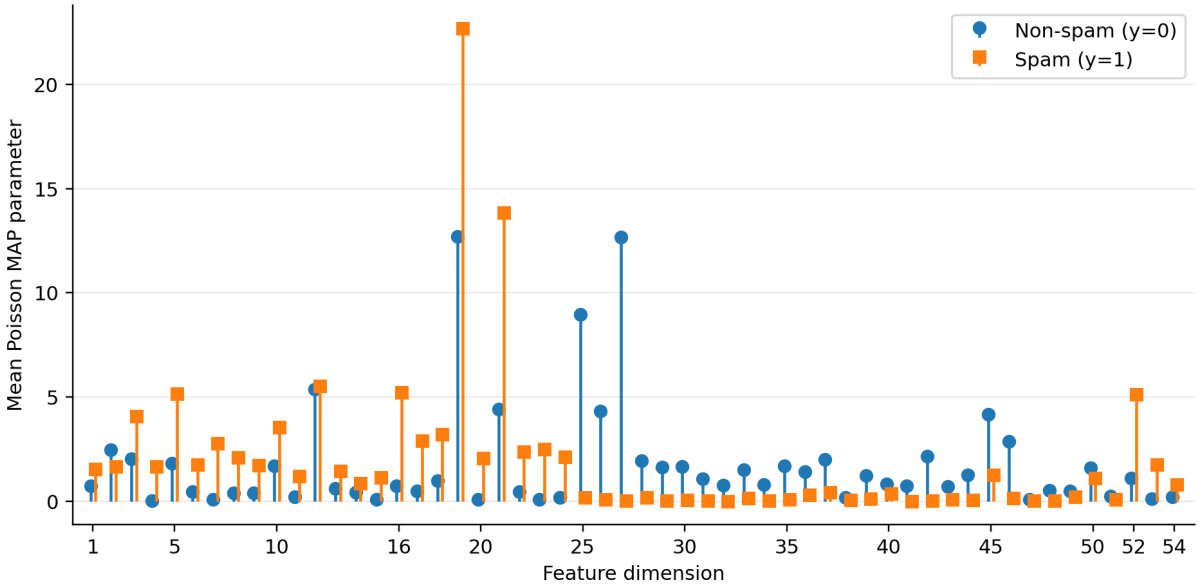


Figure 1: Mean Poisson MAP parameters for both classes. Stems are shifted slightly to distinguish the classes.

Dimension 16 is the word “free” and dimension 52 is “!”. Their mean rates for non-spam and spam are, respectively,

Feature	Non-spam	Spam
16: free	0.73960	5.21371
52: !	1.09615	5.12924

Both rates are larger for spam, so higher values of either feature favor spam through a positive contribution to the log score difference.

(c) Logistic regression by steepest ascent

Map each label 0 to -1 and retain $+1$. Append a constant $+1$ to each row, giving $a_i \in \mathbb{R}^{55}$. For a training fold, the sum log likelihood and its gradient are

$$\mathcal{L}(w) = \sum_{i \in \mathcal{T}} \log \operatorname{sigmoid}(y_i a_i^T w) = -\sum_{i \in \mathcal{T}} \log(1 + e^{-y_i a_i^T w}),$$

$$\nabla \mathcal{L}(w) = \sum_{i \in \mathcal{T}} y_i a_i \operatorname{sigmoid}(-y_i a_i^T w), \qquad \operatorname{sigmoid}(z) = \frac{1}{1 + e^{-z}}.$$

Starting at $w^{(0)} = 0$, perform exactly 1,000 updates

$$w^{(t)} = w^{(t-1)} + \eta \nabla \mathcal{L}(w^{(t-1)}), \qquad \eta = \frac{0.01}{4600} \approx 2.173913 \times 10^{-6}.$$

The denominator is 4,600 as specified, even though a training fold has 4,140 rows. No penalty is added.

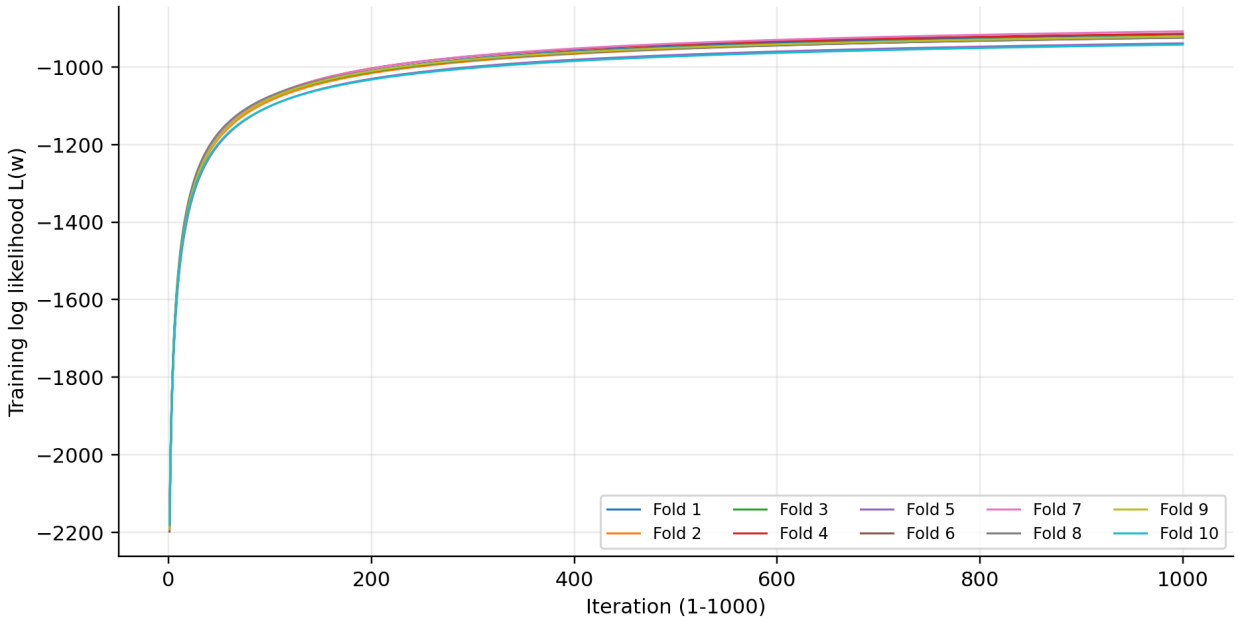


Figure 2: Training log likelihood after each steepest-ascent update, for all ten training runs in one figure.

Every run starts at $-4140 \log 2 = -2869.62933$. The curves increase toward higher likelihood. After 1,000 steps, the objectives range from -942.37473 to -908.17458. The fixed small step produces gradual improvement.

(d) Newton's method: derivation and implementation

Write $g_t = \nabla\mathcal{L}(w_t)$ and $H_t = \nabla^2\mathcal{L}(w_t)$. For the quadratic approximation

$$Q_t(w) = \mathcal{L}(w_t) + (w - w_t)^T g_t + \tfrac{1}{2}(w - w_t)^T H_t(w - w_t),$$

the stationarity equation is

$$\nabla Q_t(w) = g_t + H_t(w - w_t) = 0 \quad \Longrightarrow \quad w_{t+1} = w_t - H_t^{-1} g_t.$$

For logistic regression, let $q_i = \text{sigmoid}(y_i a_i^T w_t)$ and $R_t = \text{diag}(q_i(1 - q_i))$. Differentiating the gradient gives

$$H_t = -\sum_{i \in \mathcal{T}} a_i a_i^T q_i(1 - q_i) = -A^T R_t A.$$

The Hessian is negative semidefinite, so a well-defined stationary point maximizes the approximation. Substitution gives

$$w_{t+1} = w_t + (A^T R_t A)^{-1} A^T [y \odot \text{sigmoid}(-y \odot A w_t)].$$

The implementation solves $(A^T R_t A)\Delta_t = g_t$ and sets $w_{t+1} = w_t + \Delta_t$, using a least-squares pseudoinverse solve rather than an explicit inverse. Starting at zero, it performs all 100 unregularized full Newton updates for every fold.

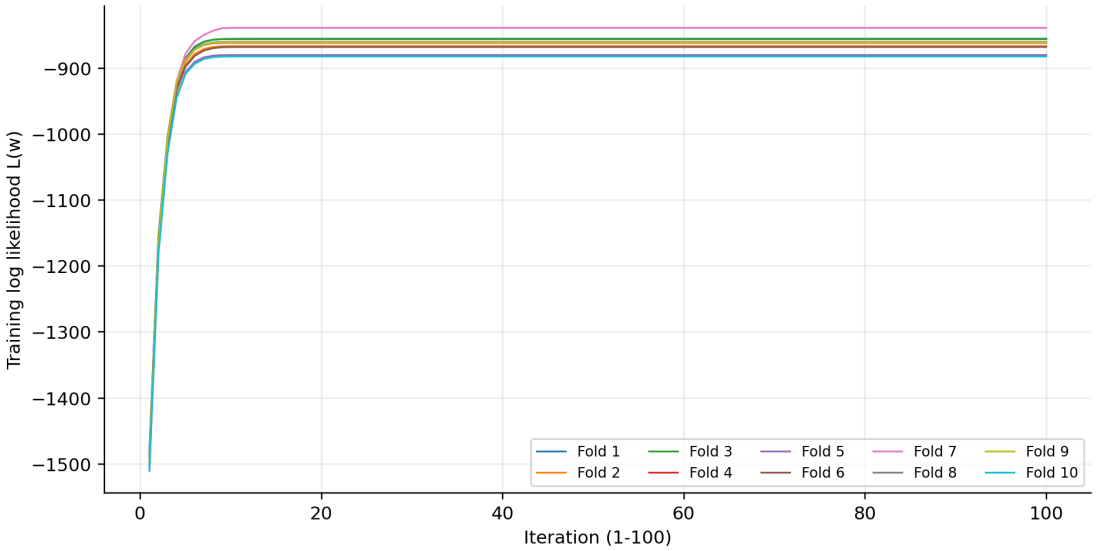


Figure 3: Training log likelihood after each Newton update, for all ten training runs in one figure.

(e) Newton logistic regression test results

Predict spam if $a^T w \geq 0$, equivalently $\text{sigmoid}(a^T w) \geq 1/2$.

Ground truth	Predicted 0	Predicted 1
0 (non-spam)	2647	140
1 (spam)	220	1593

$$\text{Accuracy} = \frac{2647 + 1593}{4600} = 92.1739\%.$$

The entries sum to 4,600. Newton's method reaches higher training likelihood than the 1,000 steepest-ascent steps and makes 230 fewer held-out mistakes than naive Bayes on these folds.

Problem 3: AdaBoost with bootstrap least squares

The toy data contain $n = 1000$ rows with labels ± 1. The third feature is already one, so no intercept is appended. Initialize $D_1(i) = 1/n$ and $F_0(x) = 0$. For $t = 1, \ldots, 2500$:

1. Sample a bootstrap multiset B_t of exactly n indices with replacement from D_t and fit

$$\beta_t = \arg\min_{\beta} \sum_{j \in B_t} (y_j - x_j^T \beta)^2.$$

   Repeated sampled indices are retained as repeated rows.

2. Set $h_t(x) = \operatorname{sign}(x^T \beta_t)$, assigning zero to $+1$, and evaluate on the original training set:

$$\epsilon_t = \sum_i D_t(i) \mathbf{1}\{h_t(x_i) \neq y_i\}.$$

   If $\epsilon_t > 1/2$, negate the whole β_t, including its offset, and recompute the error.

3. Update the ensemble and sampling distribution:

$$\alpha_t = \frac{1}{2} \log \frac{1 - \epsilon_t}{\epsilon_t}, \qquad F_t(x) = F_{t-1}(x) + \alpha_t h_t(x),$$
$$D_{t+1}(i) = \frac{D_t(i) e^{-\alpha_t y_i h_t(x_i)}}{Z_t}, \qquad Z_t = \sum_i D_t(i) e^{-\alpha_t y_i h_t(x_i)}.$$

The boosted prediction is $\operatorname{sign} F_t(x)$, with zero assigned to $+1$. Sampling uses seed 4721. If $\epsilon_t = 1/2$, then $\alpha_t = 0$ and the ensemble is unchanged; all 2,500 rounds are retained.

(a) Empirical error and upper bounds

The mistake indicator satisfies $\mathbf{1}\{f_{\text{boost}}^{(t)}(x_i) \neq y_i\} \leq e^{-y_i F_t(x_i)}$. Hence

$$\begin{aligned} \widehat{R}_t &= \frac{1}{n} \sum_i \mathbf{1}\{f_{\text{boost}}^{(t)}(x_i) \neq y_i\} \\ &\leq \frac{1}{n} \sum_i e^{-y_i F_t(x_i)} = \prod_{s=1}^{t} Z_s = \prod_{s=1}^{t} 2\sqrt{\epsilon_s(1 - \epsilon_s)}. \end{aligned}$$

Unrolling the normalized distribution updates from $D_1(i) = 1/n$ yields the product identity. The selected coefficient gives

$$Z_s = (1 - \epsilon_s) e^{-\alpha_s} + \epsilon_s e^{\alpha_s} = 2\sqrt{\epsilon_s(1 - \epsilon_s)}.$$

Writing $\gamma_s = 1/2 - \epsilon_s$ and applying $\log(1 - u) \leq -u$ gives the common looser bound

$$\widehat{R}_t \leq \prod_{s=1}^{t} \sqrt{1 - 4\gamma_s^2} \leq \exp\left(-2 \sum_{s=1}^{t} \gamma_s^2 \right).$$

Both upper bounds are plotted with the empirical error.

(a) Training error and upper bounds

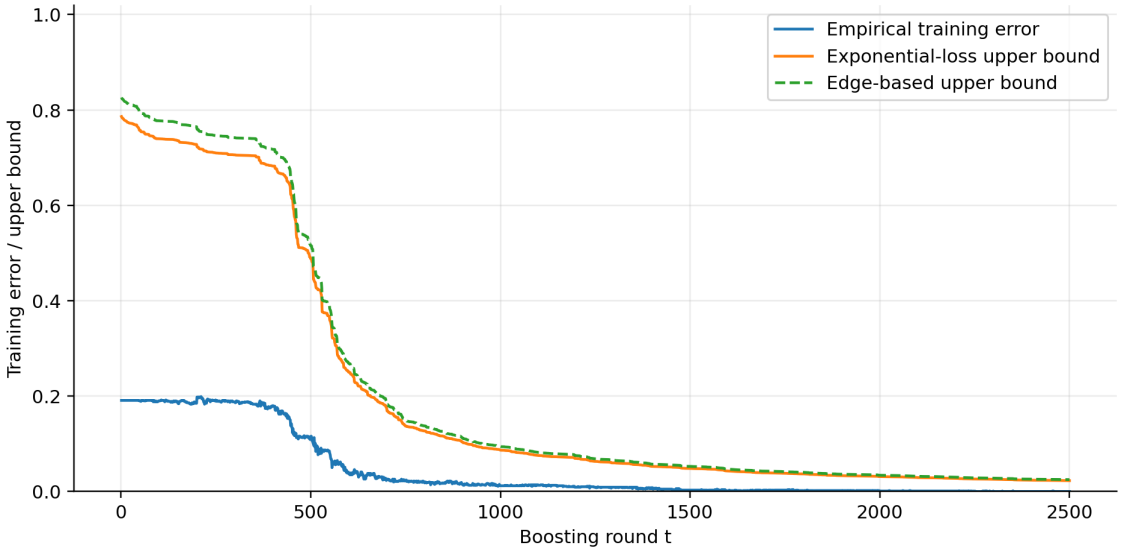


Figure 4: Training error, exact exponential-loss bound, and looser edge-based bound over all 2,500 rounds.

The final classifier makes 0/1000 training mistakes. At round 2,500 the exponential-loss bound is 0.0227164 and the edge-based bound is 0.0248033. The bounds hold at every round; the empirical error itself need not decrease at every round. These are training results, not estimates of test error.

(b) Average distribution on the data

Average the probabilities used for bootstrap sampling:

$$\overline{D}(i) = \frac{1}{2500} \sum_{t=1}^{2500} D_t(i).$$

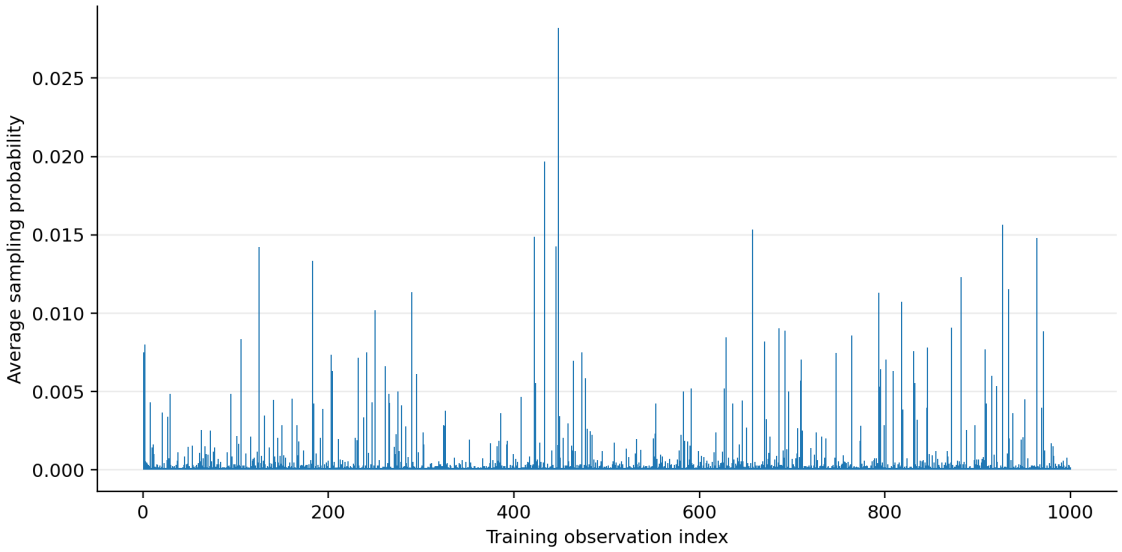


Figure 5: Stem plot of the average sampling distribution over all 1,000 observations.

The probabilities sum to one and concentrate on repeatedly difficult observations. Observation 448 has the largest mean probability, approximately 0.02820, compared with 0.001 initially.

(c) Weak-classifier errors and ensemble coefficients

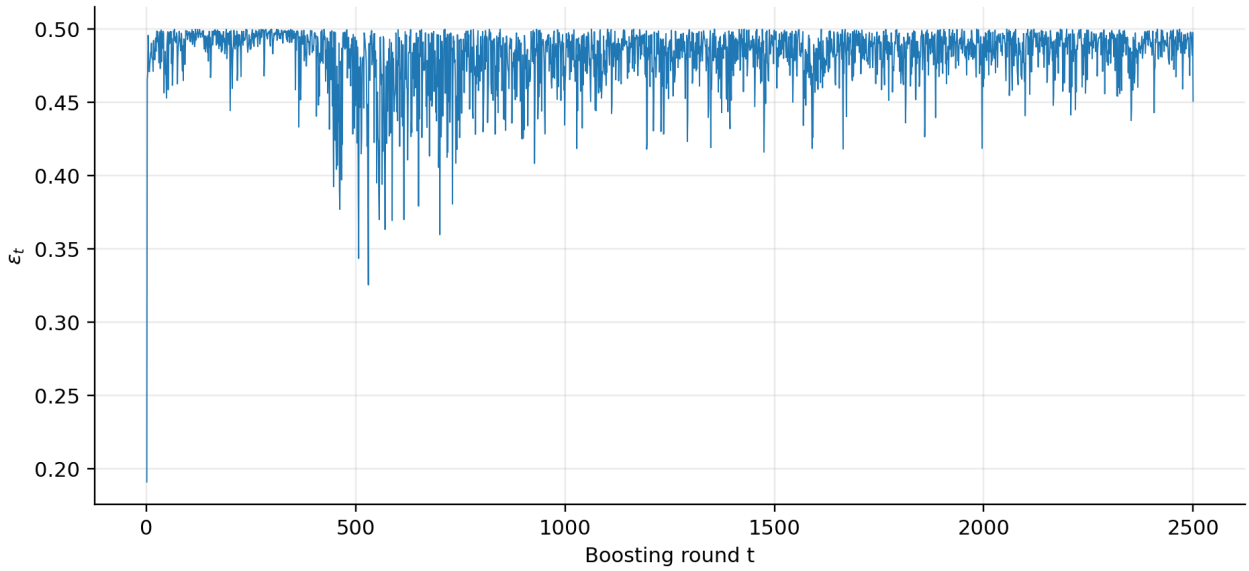


Figure 6: Weighted weak-classifier error ϵ_t, after sign reversals, over 2,500 rounds.

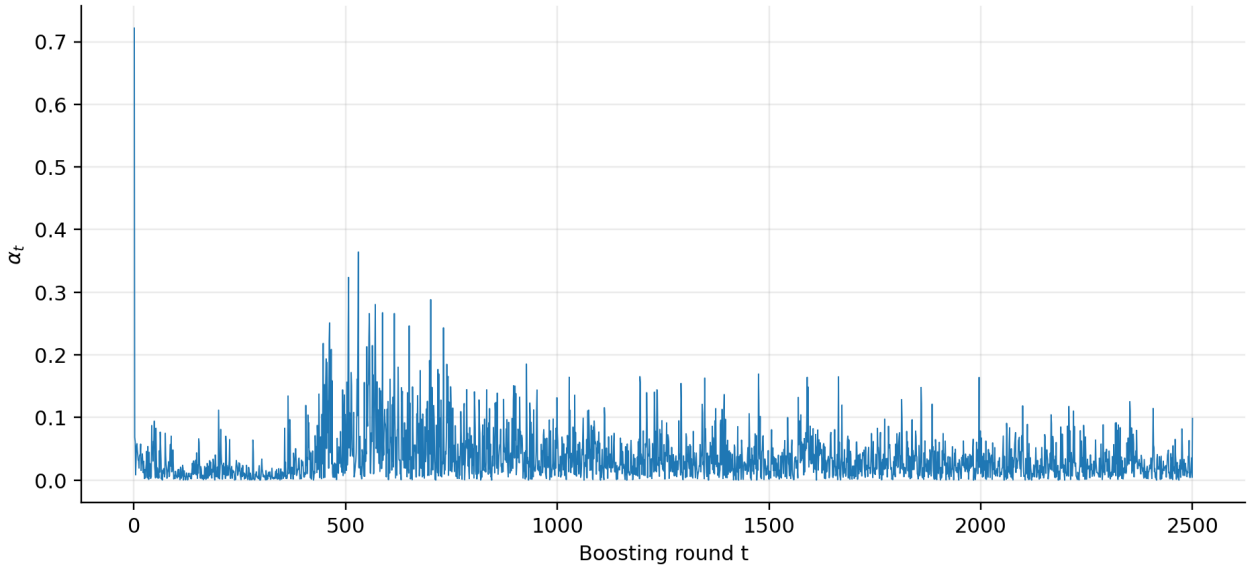


Figure 7: AdaBoost coefficient $\alpha_t = \frac{1}{2}\log[(1-\epsilon_t)/\epsilon_t]$, shown in a separate figure.

The errors range from 0.191 to 0.5; coefficients range from 0 to 0.72176. Smaller weighted errors receive larger coefficients, while errors close to chance receive small coefficients. Exactly $\epsilon_t = 1/2$ gives $\alpha_t = 0$. Sign reversal was applied in 935 rounds before recording errors and coefficients. The variation reflects the changing distribution and the random bootstrap samples.